

Лабораторная работа №3

Дисциплина: Администрирование сетевых подсистем

Бахи сиди али темассини

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1. Цель работы
2. Выполнение лабораторной работы
3. Выводы

Раздел 1

1. Цель работы

1.1 Цель работы

- Приобретение практических навыков установки и конфигурирования DHCP-сервера

Раздел 2

2. Выполнение лабораторной работы

2.1 Установка DHCP-сервера

- Переход в режим суперпользователя
- Установка DHCP-сервера Kea через dnf

```
[root@server.bahis.net ~]# dnf -y install kea
Rocky Linux 9 - BaseOS                               3.6 MB/s | 12 MB   00:03
Rocky Linux 9 - AppStream                             3.8 MB/s | 15 MB   00:03
Rocky Linux 9 - Extras                               34 kB/s | 17 kB    00:00
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
kea                                    x86_64            2.6.4-1.el9       epel               1.3 M
Installing dependencies:
kea-libs                             x86_64            2.6.4-1.el9       epel               3.0 M
log4cplus                            x86_64            2.0.5-15.el9      epel               337 k
mariadb-connector-c                   x86_64            3.2.6-1.el9_0     appstream          195 k
mariadb-connector-c-config            noarch            3.2.6-1.el9_0     appstream          9.8 k
postgresql-private-libs               x86_64            13.23-1.el9_7     appstream          140 k
Transaction Summary
=====
Install 6 Packages

Total download size: 5.0 M
Installed size: 18 M
Downloading Packages:
(1/6): log4cplus-2.0.5-15.el9.x86_64.rpm               607 kB/s | 337 kB   00:00
(2/6): mariadb-connector-c-3.2.6-1.el9_0.x86_64.rpm     771 kB/s | 195 kB   00:00
(3/6): kea-2.6.4-1.el9.x86_64.rpm                     1.5 MB/s | 1.3 MB   00:00
(4/6): mariadb-connector-c-config-3.2.6-1.el9_0.noarch.rpm 159 kB/s | 9.8 kB   00:00
```

Рисунок 1: Установка пакета kea через dnf

2.2 Конфигурирование DHCP-сервера

- Отредактирован `/etc/kea/kea-dhcp4.conf`
- Указан интерфейс `eth1`
- Настроен DNS `192.168.1.1`
- Задан домен `bahis.net`
- Определена подсеть `192.168.1.0/24`
- Диапазон `192.168.1.30–192.168.1.199`
- Указан шлюз `192.168.1.1`

```
[root@server.bahis.net ~]# cat /etc/kea/kea-dhcp4.conf
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "eth1" ]
    },
    "option-data": [
      {
        "name": "domain-name-servers",
        "data": "192.168.1.1"
      },
      {
        "code": 15,
        "data": "bahis.net"
      },
      {
        "name": "domain-search",
```

- Выполнена проверка kea-dhcp4 -t
- Ошибки отсутствуют
- Прослушивание на eth1 активно

```

root@server.bahis.net ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-02-09 18:43:23.465 INFO [kea-dhcp4.hosts/11223.140569492961408] HOSTS_BACKENDS_REGISTERED the following
; host backend types are available: mysql postgresql
2026-02-09 18:43:23.472 WARN [kea-dhcp4.dhcp4/11223.140569492961408] DHCP4_MT_DISABLED_QUEUE_CONTROL di
; enabling dhcp queue control when multi-threading is enabled.
2026-02-09 18:43:23.472 WARN [kea-dhcp4.dhcp4/11223.140569492961408] DHCP4_RESERVATIONS_LOOKUP_FIRST_ENABLE
; Multi-threading is enabled and host reservations lookup is always performed first.
2026-02-09 18:43:23.473 INFO [kea-dhcp4.dhcp4/11223.140569492961408] DHCP4_CFGMGR_NEW_SUBNET4 a new sub
; net has been added to configuration: 192.168.1.0/24 with params: valid-lifetime=7200
2026-02-09 18:43:23.473 INFO [kea-dhcp4.dhcp4/11223.140569492961408] DHCP4_CFGMGR_SOCKET_TYPE_SELECT us
; ing socket type raw
2026-02-09 18:43:23.474 INFO [kea-dhcp4.dhcp4/11223.140569492961408] DHCP4_CFGMGR_ADD_IFACE listening o
; n interface eth1
2026-02-09 18:43:23.474 INFO [kea-dhcp4.dhcp4/11223.140569492961408] DHCP4_CFGMGR_SOCKET_TYPE_DEFAULT "
; dhcp-socket-type" not specified , using default socket type raw
root@server.bahis.net ~]# systemctl --system daemon-reload
root@server.bahis.net ~]# systemctl enable kea-dhcp4.service
Created symlink /etc/systemd/system/multi-user.target.wants/kea-dhcp4.service → /usr/lib/systemd/system/kea-

```

Рисунок 3: Проверка конфигурации kea-dhcp4 -t

- В прямую зону добавлена A-запись `dhcp.bahis.net`
- Обновлён серийный номер зоны

```
STTL 1D
@ IN SOA @ server.bahis.net. (
    2026020919 ; serial
    1D ; refresh
    1H ; retry
    1W ; expire
    3H ) ; minimum
NS @
A 192.168.1.1
ORIGIN bahis.net.
server A 192.168.1.1
ns A 192.168.1.1
dhcp A 192.168.1.1
```

- В обратную зону добавлена PTR-запись
- Связь 192.168.1.1 → dhcp.bahis.net
- Обновлён серийный номер

```
$TTL 1D
@      IN SOA  @ server.bahis.net. (
        2026020919      ; serial
        1D              ; refresh
        1H              ; retry
        1W              ; expire
        3H )            ; minimum
      NS      @
      A       192.168.1.1
      PTR     server.bahis.net.
$ORIGIN 1.168.192.in-addr.arpa.
1       PTR   server.bahis.net.
1       PTR   ns.bahis.net.
1       PTR   dhcp.bahis.net.
```

- Перезапущена служба named
- Проверено разрешение dhcp.bahis.net
- Потерь пакетов нет

```
root@server.bahis.net ~]# systemctl restart named
root@server.bahis.net ~]# ping dhcp.bahis.net
PING dhcp.bahis.net (192.168.1.1) 56(84) bytes of data.
64 bytes from server.bahis.net (192.168.1.1): icmp_seq=1 ttl=64 time=0.016 ms
64 bytes from server.bahis.net (192.168.1.1): icmp_seq=2 ttl=64 time=0.037 ms
64 bytes from dhcp.bahis.net (192.168.1.1): icmp_seq=3 ttl=64 time=0.047 ms
64 bytes from dhcp.bahis.net (192.168.1.1): icmp_seq=4 ttl=64 time=0.063 ms
64 bytes from ns.bahis.net (192.168.1.1): icmp_seq=5 ttl=64 time=0.031 ms
64 bytes from server.bahis.net (192.168.1.1): icmp_seq=6 ttl=64 time=0.036 ms
64 bytes from ns.bahis.net (192.168.1.1): icmp_seq=7 ttl=64 time=0.037 ms
```

Рисунок 6: Проверка разрешения имени dhcp.bahis.net командой ping

- В firewalld добавлена служба dhcpd
- Правило добавлено в текущую и постоянную конфигурацию
- Восстановлены SELinux-контексты

```

root@server.bahis.net ~]# chown -R named:named /var/named
root@server.bahis.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns ssh
root@server.bahis.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit aus
bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bi
estnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit
condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dr
els docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman
n-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-cl
glia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-cl
ec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell k
kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-
-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-reado
let-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp m
eve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd neb
bios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-s
console ovirt-vmconsole plex pmcd pmpoxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus
eus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis r
htinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips s
smtp-submission smtps snmp snmptls snmptls-lsra snmptrap spideroak-lansync spotify-sync squid ssdp s
n-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentac
tile38 tinc tor-socks transmission-client upnp-client vdsu vnc-server warpinator wbem-http wbem-https
ward ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh
ient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
root@server.bahis.net ~]# firewall-cmd --add-service=dhcp
success
root@server.bahis.net ~]# firewall-cmd --add-service=dhcp --permanent
success
root@server.bahis.net ~]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
root@server.bahis.net ~]# restorecon -vR /var/named
root@server.bahis.net ~]# restorecon -vR /var/lib/kea/
root@server.bahis.net ~]# tail -f /var/log/messages
Feb 9 18:48:53 server named[11302]: configuring command channel from '/etc/rndc.key'

```

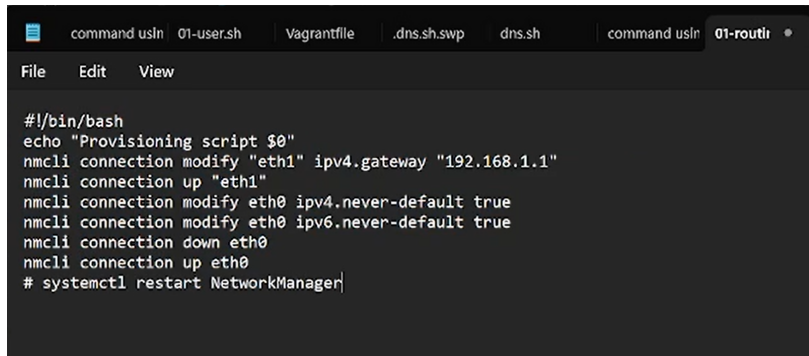

- Выполнен мониторинг /var/log/messages
- Запущена служба kea-dhcp4.service
- Состояние — рабочее

```
[root@server.bahis.net ~]# restorecon -vR /var/named
[root@server.bahis.net ~]# restorecon -vR /var/lib/kea/
[root@server.bahis.net ~]# tail -f /var/log/messages
Feb  9 18:48:53 server named[11302]: configuring command channel from '/etc/rndc.key'
Feb  9 18:48:53 server named[11302]: command channel listening on 127.0.0.1#953
Feb  9 18:48:53 server named[11302]: configuring command channel from '/etc/rndc.key'
Feb  9 18:48:53 server named[11302]: command channel listening on ::1#953
Feb  9 18:48:53 server named[11302]: managed-keys-zone: loaded serial 3
Feb  9 18:48:53 server named[11302]: zone 1.168.192.in-addr.arpa/IN: loaded serial 2026020920
Feb  9 18:48:53 server named[11302]: zone bahis.net/IN: loaded serial 2026020919
Feb  9 18:48:53 server named[11302]: all zones loaded
Feb  9 18:48:53 server systemd[1]: Started Berkeley Internet Name Domain (DNS).
Feb  9 18:48:53 server named[11302]: running
```

Рисунок 8: Мониторинг журнала и запуск kea-dhcp4.service

2.3 Анализ работы DHCP-сервера

- Создан скрипт 01-routing.sh
- Настроен шлюз 192.168.1.1 для eth1
- Отключён eth0 как default-route

A screenshot of a terminal window with a dark background. The window has several tabs at the top: 'command usin', '01-user.sh', 'Vagrantfile', '.dns.sh.swp', 'dns.sh', 'command usin', and '01-routi'. The '01-routi' tab is active. Below the tabs is a menu bar with 'File', 'Edit', and 'View'. The main area of the terminal displays the following script content:

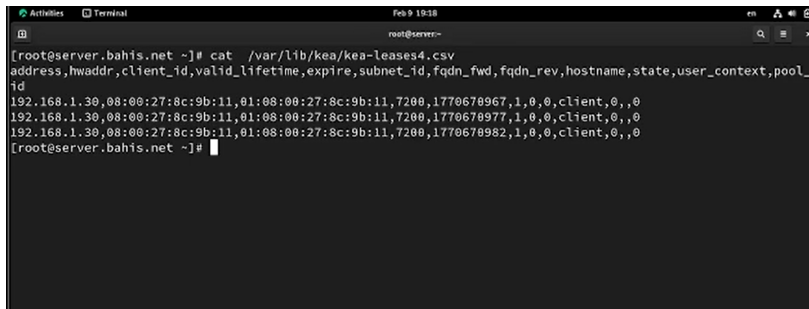
```
#!/bin/bash
echo "Provisioning script $0"
nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
nmcli connection up "eth1"
nmcli connection modify eth0 ipv4.never-default true
nmcli connection modify eth0 ipv6.never-default true
nmcli connection down eth0
nmcli connection up eth0
# systemctl restart NetworkManager
```

Рисунок 9: Скрипт 01-routing.sh для настройки маршрутизации клиента

- Скрипт подключён в Vagrantfile
- Параметр run: "always"

```
## Client configuration
config.vm.define "client", autostart: false do |client|
  client.vm.box = "rocky9"
  client.vm.hostname = 'client'
  client.vm.boot_timeout = 1440
  client.ssh.insert_key = false
  client.ssh.username = 'vagrant'
  client.ssh.password = 'vagrant'
  client.vm.network :private_network,
    type: "dhcp",
  virtualbox____intnet: true
  client.vm.provision "client dummy",
    type: "shell",
    preserve_order: true,
    path: "provision/client/01-dummy.sh"
  client.vm.provision "client routing",
    type: "shell",
    preserve_order: true,
    run: "always",
    path: "provision/client/01-routing.sh"
  client.vm.provider :virtualbox do |v|
    v.linked_clone = true
  end
  # Customize the amount of memory on the VM
```

- В `/var/lib/kea/kea-leases4.csv` зафиксирована аренда `192.168.1.30`
- MAC клиента `08:00:27:ef:c7:e6`
- Время аренды 7200 секунд
- Подсеть ID 1
- Имя клиента `client`



```
root@server:~  
[root@server.bahis.net ~]# cat /var/lib/kea/kea-leases4.csv  
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_  
id  
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670967,1,0,0,client,0,,0  
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670977,1,0,0,client,0,,0  
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670982,1,0,0,client,0,,0  
[root@server.bahis.net ~]#
```

Рисунок 11: Файл `kea-leases4.csv` с выданными арендами

- Выполнена команда `ifconfig` на клиенте
- `eth0` – 10.0.2.15/24 (NAT)
- `eth1` – 192.168.1.30/24 (DHCP)
- Подтверждён MAC-адрес

```
[root@client.bahis.net ~]# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe98:fdfe prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe98:fdfe prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:98:fd:fe txqueuelen 1000 (Ethernet)
    RX packets 1457 bytes 165554 (161.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1259 bytes 196446 (191.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.30 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe8c:9b11 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8c:9b:11 txqueuelen 1000 (Ethernet)
    RX packets 430 bytes 42010 (41.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1066 bytes 94211 (92.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 17 bytes 2045 (1.9 KiB)
```

- Повторная проверка leases
- Аренда активна
- DHCP работает корректно

```
[root@server.bahis.net ~]# cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool
id
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670967,1,0,0,client,0,,0
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670977,1,0,0,client,0,,0
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670982,1,0,0,client,0,,0
[root@server.bahis.net ~]# cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool
id
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670967,1,0,0,client,0,,0
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670977,1,0,0,client,0,,0
192.168.1.30,08:00:27:8c:9b:11,01:08:00:27:8c:9b:11,7200,1770670982,1,0,0,client,0,,0
[root@server.bahis.net ~]#
```

Рисунок 13: Повторная проверка списка выданных адресов

2.4 Настройка обновления DNS-зоны

- Создан каталог `/etc/named/keys`
- Сгенерирован TSIG-ключ DHCP_UPDATER (HMAC-SHA512)
- Настроены права доступа

```
[root@server.bahis.net ~]# mkdir -p /etc/named/keys
[root@server.bahis.net ~]# tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key
[root@server.bahis.net ~]# nano /etc/named/keys/dhcp_updater.key
[root@server.bahis.net ~]# chown -R named:named /etc/named/keys
[root@server.bahis.net ~]#
```

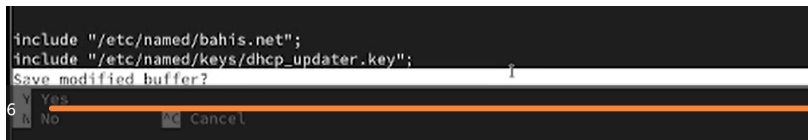
Рисунок 14: Создание TSIG-ключа и настройка прав доступа

- В файле `dhcр_updater.key` указаны имя, алгоритм и секрет

```
xkey "DHCP_UPDATER" {  
    algorithm hmac-sha512;  
    secret "s404eVs5HYepkrt1luyN0l3qyPQH4tavBm2JF0D2ZIqFFpPSeIoJrhfJMY0n7rc4EQyceTkTD5kuazmGmrZ4qg==";  
};
```

Рисунок 15: Содержимое файла `dhcр_updater.key`

- В `named.conf` добавлена директива `include` для ключа



```
include "/etc/named/bahis.net";  
include "/etc/named/keys/dhcp_updater.key";  
Save modified buffer?  
6 Y Yes  
  N No      [OK] Cancel
```

Рисунок 16: Подключение TSIG-ключа в `named.conf`

- В зонах добавлена update-policy
- Разрешено обновление A- и PTR-записей
- Используется механизм DHCID

```
zone "bahis.net" IN {  
    type master;  
    file "master/fz/bahis.net";  
    update-policy {  
        grant DHCP_UPDATER wildcard *.bahis.net A DHCID;  
    };  
};  
  
zone "1.168.192.in-addr.arpa" IN {  
    type master;  
    file "master/rz/192.168.1";  
    update-policy {  
        grant DHCP_UPDATER wildcard *.1.168.192.in-addr.arpa PTR DHCID;  
    };  
};
```

Рисунок 17: Настройка update-policy для прямой и обратной зоны

- Проверка `named-checkconf`
- Перезапуск `named`
- Ошибки отсутствуют

```
[root@server.bahis.net ~]# named-checkconf  
[root@server.bahis.net ~]# systemctl restart named
```

Рисунок 18: Проверка конфигурации и перезапуск `named`

- Создан `/etc/kea/tsig-keys.json`
- Перенесён секретный ключ
- Владелец `kea:kea`, права `640`

```
[root@server.bahis.net ~]# touch /etc/kea/tsig-keys.json
[root@server.bahis.net ~]# nano /etc/kea/tsig-keys.json
[root@server.bahis.net ~]# nano /etc/named/keys/dhcp_updater.key
[root@server.bahis.net ~]# nano /etc/kea/tsig-keys.json
[root@server.bahis.net ~]# chown kea:kea /etc/kea/tsig-keys.json
[root@server.bahis.net ~]# chmod 640 /etc/kea/tsig-keys.json
[root@server.bahis.net ~]#
```

Рисунок 19: Создание и настройка файла `tsig-keys.json` для Kea

- В файле описан ключ DHCP_UPDATER
- Алгоритм hmac-sha512

```
'tsig-keys': [  
  {  
    'name': "DHCP_UPDATER",  
    'algorithm': "hmac-sha512",  
    'secret': 'psFFSGdUIwK36l1G82L0w15PuIH4W5uB8h/cw7F0XDsjniBbwm/59tM+V3ydcCs15VLpe2pxlUlg'  
  },  
]
```

Рисунок 20: Содержимое файла tsig-keys.json

- Настроен /etc/kea/kea-dhcp-ddns.conf
- Адрес 127.0.0.1, порт 53001
- Подключён tsig-keys.json
- Указаны зоны bahis.net. и 1.168.192.in-addr.arpa.

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "eth1" ]
    },
    "option-data": [
      {
        "name": "domain-name-servers",
        "data": "192.168.1.1"
      },
      {
        "code": 15,
        "data": "bahis.net"
      },
      {
        "name": "domain-search",
        "data": "bahis.net"
      }
    ]
  },
  "subnet4": [
    {
      "id": 1,
      "subnet": "192.168.1.0/24",
      "pools": [
        { "pool": "192.168.1.30 - 192.168.1.199" }
      ],
      "option-data": [
        {
          "name": "routers",
          "data": "192.168.1.1"
        },
        {
          "name": "domain-name-servers",
          "data": "192.168.1.1"
        },
        {
          "code": 15,
          "data": "bahis.net"
        },
        {
          "name": "domain-search",
          "data": "bahis.net"
        }
      ]
    }
  ]
}
```

- Проверка kea-dhcp-ddns -t
- Служба запущена
- Статус active (running)

```

root@server.bahis.net ~]# chown kea:kea /etc/kea/kea-dhcp-ddns.conf
root@server.bahis.net ~]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
2026-02-09 20:52:42.691 INFO [kea-dhcp-ddns.dctl/11521.140179518173312] DCTL_CONFIG_CHECK_COMPLETE server
has completed configuration check: listening on 127.0.0.1, port 53001, using UDP, result: success(0), test
configuration check successful
root@server.bahis.net ~]# systemctl enable --now kea-dhcp-ddns.service
root@server.bahis.net ~]# systemctl status kea-dhcp-ddns.service
kea-dhcp-ddns.service - Kea DHCP-DDNS Server
Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disabled)
Active: active (running) since Mon 2026-02-09 20:54:03 UTC; 10s ago
Docs: man:kea-dhcp-ddns(8)
Main PID: 11556 (kea-dhcp-ddns)
Tasks: 5 (limit: 4498)
Memory: 1.9M (peak: 2.1M)
CPU: 10ms
CGroup: /system.slice/kea-dhcp-ddns.service
└─11556 /usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf

Feb 09 20:54:03 server.bahis.net systemd[1]: Started Kea DHCP-DDNS Server.
Feb 09 20:54:03 server.bahis.net kea-dhcp-ddns[11556]: 2026-02-09 20:54:03.252 INFO [kea-dhcp-ddns.dctl/
Feb 09 20:54:03 server.bahis.net kea-dhcp-ddns[11556]: INFO COMMAND_ACCEPTOR_START Starting to accept co
Feb 09 20:54:03 server.bahis.net kea-dhcp-ddns[11556]: INFO DCTL_CONFIG_COMPLETE server has completed co
Feb 09 20:54:03 server.bahis.net kea-dhcp-ddns[11556]: INFO DHCP_DDNS_STARTED Kea DHCP-DDNS server versi

```

Рисунок 22: Проверка конфигурации и запуск kea-dhcp-ddns

- В kea-dhcp4.conf включены:
- "enable-updates": true
- "ddns-qualifying-suffix": "bahis.net"
- "ddns-override-client-update": true

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "eth1" ]
    },
    "dhcp-ddns": {
      "enable-updates": true
    },
    "ddns-qualifying-suffix": "user.net",
    "ddns-override-client-update": true,
    "option-data": [
      {
        "name": "domain-name-servers",
        "data": "192.168.1.1"
      }
    ]
  }
}
```

Рисунок 23: Включение динамических обновлений в kea-dhcp4.conf

- Проверка kea-dhcp4 -t
- Перезапуск kea-dhcp4.service
- Статус active (running)

```

dhcp socket type not specified, using default socket type raw
[root@server.bahis.net ~]# systemctl restart kea-dhcp4.service
bash: systemctl: command not found...
[root@server.bahis.net ~]# systemctl restart kea-dhcp4.service
[root@server.bahis.net ~]# systemctl status kea-dhcp4.service
● kea-dhcp4.service - Kea DHCPv4 Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-02-09 20:59:55 UTC; 17s ago
     Docs: man:kea-dhcp4(8)
  Main PID: 11591 (kea-dhcp4)
    Tasks: 6 (limit: 4498)
   Memory: 6.8M (peak: 7.0M)
      CPU: 25ms
    CGroup: /system.slice/kea-dhcp4.service
            └─11591 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.689 INFO [kea-dhcp4.dhcp4/11591.1
Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.700 INFO [kea-dhcp4.dhcp4srv/11591
Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.701 INFO [kea-dhcp4.dhcp4srv/11591
Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.701 INFO [kea-dhcp4.dhcp4srv/11591
Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.701 INFO [kea-dhcp4.dhcp4srv/11591
Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.701 INFO [kea-dhcp4.dhcp4srv/11591
Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.701 INFO [kea-dhcp4.dhcp4srv/11591
Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.713 INFO [kea-dhcp4.dhcp4srv/11591
Feb 09 20:59:55 server.bahis.net kea-dhcp4[11591]: 2026-02-09 20:59:55.713 WARN [kea-dhcp4.dhcp4/11591.1

```

Рисунок 24: Проверка и перезапуск kea-dhcp4

- На клиенте выполнено переполучение IP (`nmcli connection down/up`)
- Инициировано динамическое обновление DNS

```
[root@client.bahis.net ~]# nmcli connection down eth1
Connection 'eth1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/16)
[root@client.bahis.net ~]# nmcli connection up eth1
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/17)
[root@client.bahis.net ~]#
```

Рисунок 25: Переполучение IP-адреса на клиенте

2.5 Анализ работы DHCP-сервера после настройки обновления DNS-зоны

- Выполнен `dig @192.168.1.1 client.bahis.net`
- Статус NOERROR
- Флаг aa — сервер авторитетный
- Получена запись A 192.168.1.30
- TTL 2400 секунд

```
[root@client.bahis.net ~]# dig @192.168.1.1 client.bahis.net

; <<>> DiG 9.16.23-RH <<>> @192.168.1.1 client.bahis.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 22992
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 046a93e86a549cf101000000698b362243443cd1554b84f3 (good)
;; QUESTION SECTION:
;client.bahis.net.                IN      A

;; AUTHORITY SECTION:
bahis.net.                10800   IN      SOA     bahis.net. server.bahis.net. 2026020919
```

2.6 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Создан каталог `/vagrant/provision/server/dhcp/etc/kea`
- Скопированы конфигурации DHCP и DNS

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/dhcp/etc/kea
[root@server.bahis.net server]# cp -R /etc/kea/* /vagrant/provision/server/dhcp/etc/kea/
[root@server.bahis.net server]# cd /vagrant/provision/server/dns/
[root@server.bahis.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
[root@server.bahis.net dns]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/
cp: overwrite '/vagrant/provision/server/dns/etc/named/bahis.net'? y
cp: overwrite '/vagrant/provision/server/dns/etc/named/user.net'? y
[root@server.bahis.net dns]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch dhcp.sh
[root@server.bahis.net server]# chmod +x dhcp.sh
[root@server.bahis.net server]# touch dhcp.sh
```

- Создан скрипт `dhcpr.sh`
- Установка `kea`
- Копирование конфигураций
- Настройка прав доступа
- Восстановление SELinux
- Настройка `firewalld`
- Запуск `kea-dhcp4` и `kea-dhcp-ddns`

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install kea
echo "Copy configuration files"
cp -R /vagrant/provision/server/dhcp/etc/kea/* /etc/kea/
echo "Fix permissions"
chown -R kea:kea /etc/kea
chmod 640 /etc/kea/tsig-keys.json
restorecon -vR /etc
restorecon -vR /var/lib/kea
echo "Configure firewall"
firewall-cmd --add-service dhcp
firewall-cmd --add-service dhcp --permanent
```

- В Vagrantfile добавлен provision-блок
- Путь provision/server/dhcp.sh
- Сохранён порядок выполнения

```
server.vm.provision "server dhcp",  
type: "shell",  
preserve_order: true,  
path: "provision/server/dhcp.sh"  
server.vm.provider "virtualbox" do |v|
```

Рисунок 29: Подключение dhcp.sh в Vagrantfile

Раздел 3

3. Выводы

3.1 Выводы

- Установлен и настроен DHCP-сервер Kea
- Реализована выдача адресов в сети `192.168.1.0/24`
- Настроено динамическое обновление DNS через TSIG
- Подтверждено создание записи `client.bahis.net`
- Проверена корректность работы DHCP и DDNS
- Зафиксированы аренды в `kea-leases4.csv`
- Реализована автоматизация через `dhcp.sh` и `Vagrantfile`